

Sri Lanka Institute of Advanced Technological Education (SLIATE)

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Higher National Diploma in Information Technology

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AgroSmart Marketplace: A Smart Agriculture Trading Platform

IT4052 | ICT Project (Individual)

Supervisor

Ms.K.G.D.De.A Wijesinghe

N.M.T.H Senevirathna

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1. Problem Statement

The current agricultural market structure in Sri Lanka relies heavily on intermediaries, which reduces fairness, transparency, and efficiency in trading activities. Farmers are often unable to receive fair prices for their products, as intermediaries capture a significant portion of the profit, limiting farmers' income and their ability to make independent pricing decisions.

In addition, there is no proper mechanism for direct interaction between farmers and buyers. The absence of a structured negotiation process limits buyers' ability to offer prices and results in inefficient price determination compared to real-world agricultural markets.

Buyers also face difficulties in accessing accurate and real-time market price information, leading to price inconsistencies and unfair transactions. Furthermore, the lack of reliable demand-related data prevents farmers from understanding actual market demand, often resulting in oversupply or shortages of crops and causing price instability.

Moreover, there is no effective system to collect and analyze buyer activities such as searches, views, orders and offers, which could provide valuable insights into demand patterns. The absence of a review or feedback mechanism also reduces trust, as buyers cannot evaluate product quality or farmer reliability based on previous transactions.

Therefore, these limitations highlight the need for a transparent, negotiation-based, and data-driven agricultural trading system that supports direct farmer-to-buyer interaction, enables price negotiation, provides demand analysis insights, and incorporates a feedback mechanism to improve trust, market efficiency, and decision-making for both farmers and buyers.

2. Objectives and Goals

Primary Objective: To develop a smart agriculture marketplace that enables direct farmer-to-buyer trading through a negotiation-based system, while supporting smart pricing, demand analysis, and efficient decision-making.

Specific Objectives:

1. To design and develop a platform that allows farmers to publish agricultural products and sell them directly to buyers without intermediaries.
2. To implement a negotiation-based trading system where buyers can submit price offers and farmers can accept, reject, or counter those offers.
3. To incorporate a minimum price feature that allows farmers to set a base price and ensures that offers below the minimum value are restricted or rejected.
4. To implement a smart pricing mechanism that provides suggested price ranges based on market conditions and demand levels, assisting farmers in making better pricing decisions.
5. To develop a demand analysis system that uses buyer activities such as product searches, views, offers, and orders to identify high-demand and low-demand crops and display the results through a dashboard.
6. To implement a location-based matching feature that connects buyers with nearby farmers to reduce transportation costs and improve efficiency.
7. To provide a user-friendly interface for farmers and buyers to manage products, offers, and orders efficiently.
8. To ensure transparency in the trading process by providing accurate product information, pricing details, and visibility of buyer offers.
9. To incorporate a review and feedback system that allows buyers to evaluate product quality and farmer reliability, thereby improving trust within the platform.

3. Proposed Solution / Approach

3.1. Overview of the Proposed Solution

This study proposes the development of a Smart Agriculture Marketplace that facilitates direct trading between farmers and buyers without the involvement of intermediaries. The system provides a digital platform where farmers can publish their agricultural products, and buyers can search, view, and interact with available products in a transparent and efficient manner.

Unlike traditional fixed-price systems, the platform supports a negotiation-based trading approach, allowing buyers to submit price offers and farmers to make decisions based on those offers. This enhances market flexibility and better reflects real-world agricultural trading practices.

3.2. Smart Pricing Mechanism

The platform incorporates a smart pricing mechanism that suggests suitable price ranges based on current market conditions, demand levels, and available historical data. In addition, farmers are able to set a minimum price for their products, and the system ensures that offers below this minimum value are restricted or automatically rejected. This feature helps protect farmers from unfair pricing while still allowing flexibility through negotiation, enabling them to make informed pricing decisions while maintaining competitiveness in the market.

3.3. Demand Analysis Module

The system includes a demand analysis module that evaluates buyer activities such as product searches, views, offers, and completed orders. A rule-based approach is used to analyze this data and identify high-demand and low-demand agricultural products.

The results are presented through a demand analysis dashboard, providing clear insights into demand patterns. These insights enable farmers to make better production, pricing, and selling decisions, thereby reducing the risk of oversupply or shortage.

3.4. Negotiation System

A negotiation system is integrated into the platform to allow buyers to submit price offers for published products. Multiple buyers can place offers, creating a competitive environment. Farmers can review all received offers and choose to accept, reject, or provide counter-offers based on their preferences, minimum price, and market conditions.

This functionality replicates traditional agricultural market bargaining in a structured digital environment, improving price fairness, flexibility, and overall transaction outcomes.

3.5. Location-Based Matching

The platform further incorporates a location-based matching feature, which connects buyers with nearby farmers based on district or city information. This approach reduces transportation costs, improves delivery efficiency, and supports local trading within the agricultural ecosystem.

3.6. Review and Feedback System

The system also includes a review and feedback mechanism that allows buyers to provide ratings and comments after completing transactions. This feature enhances transparency and trust by enabling future buyers to evaluate product quality and farmer reliability based on previous experiences.

4. Scope and Limitations

4.1. Scope of the Project

The proposed system focuses on developing a web-based Smart Agriculture Marketplace that facilitates direct trading between farmers and buyers through a negotiation-based.

approach.The scope of the system includes the following functionalities:

- Enabling farmers to register, log in, and publish their agricultural products.
- Allowing buyers to search, view, and interact with products including submitting price offers.
- Supporting a negotiation system where buyers can offer prices and farmers can accept, reject, or counter offers.
- Incorporating a minimum price feature that allows farmers to set a base price and restricts or rejects offers below that value.
- Providing a smart pricing mechanism that suggests suitable price ranges based on market conditions and demand levels.
- Implementing a demand analysis system that uses buyer activities such as searches, views, offers, and orders to identify high-demand and low-demand crops.
- Displaying demand insights through a dashboard to support better decision-making.
- Integrating a location-based matching feature to connect buyers with nearby farmers based on district or city.
- Offering a user-friendly interface for farmers and buyers to manage products, offers, and orders efficiently.
- Incorporating a review and feedback system to improve trust by allowing buyers to rate and comment on completed transactions.

4.2. Limitations of the Project

- The smart pricing mechanism is based on a rule-based approach**,** which may not fully capture complex and dynamic real-time market conditions.
- The demand analysis system relies on buyer-generated data such as searches, views, offers, and orders, which may affect the accuracy of demand insights, especially with limited data.
- The system does not incorporate advanced machine learning or AI models**,** limiting its ability to perform predictive analytics for pricing and demand forecasting.
- External factors such as weather conditions, transportation issues, seasonal variations, and market disruptions are not considered in the current implementation.
- The system is limited to a web-based platform and does not include a dedicated mobile application.
- Online payment integration and logistics management are not included and are considered as future enhancements.
- The negotiation outcomes depend on user participation, and low user activity may affect system effectiveness in early stages.

5. Methodology

5.1. Software Development Approach

This project follows the Agile Software Development Life Cycle (SDLC) approach to design and implement the AgroSmart Marketplace. The development process includes requirement analysis, system design, implementation, testing, and evaluation, carried out iteratively to allow continuous improvement.

Initially, requirements are gathered by analyzing the problems in the current agricultural trading system. Based on these requirements, the system architecture and key features such as product publishing, negotiation, smart pricing, demand analysis, and feedback mechanisms are designed to meet user needs.

5.2. System Design

- **Frontend Layer** – Provides user interfaces for farmers and buyers to interact with the system.
- **Backend Layer** – Handles business logic such as negotiation processing, smart pricing calculation, minimum price validation, demand analysis, and location-based matching.
- **Database Layer** – Stores user information, product details, offers, orders, reviews, and user activity data required for system operations and analysis.

5.3. Data Collection and Processing

The system collects and processes data from buyer activities, including:

- Product searches
- Product views
- Price offers
- Completed orders

This data is analyzed to identify demand patterns and user behavior, which are used for demand analysis and smart pricing support.

5.4. Smart Pricing Implementation

A **rule-based algorithm** is used to implement the smart pricing mechanism. The algorithm considers factors such as market conditions, demand level, and available historical data to generate suggested price ranges. In addition, a **minimum price validation rule** is applied, where offers below the farmer-defined minimum price are automatically restricted or rejected.

5.5. Demand Analysis Method

Demand analysis is performed by calculating a demand score using buyer activity data such as searches, views, offers, and completed orders. Based on the calculated score, products are categorized into high-demand and low-demand groups. The results are displayed through a dashboard to support better decision-making.

5.6. System Implementation

The system is implemented using web technologies:

- **Frontend:** HTML, Tailwind CSS, JavaScript, Alpine.js
- **Backend:** PHP / Laravel
- **Database:** MySQL
- **Authentication**: Laravel Jetstream and Fortify

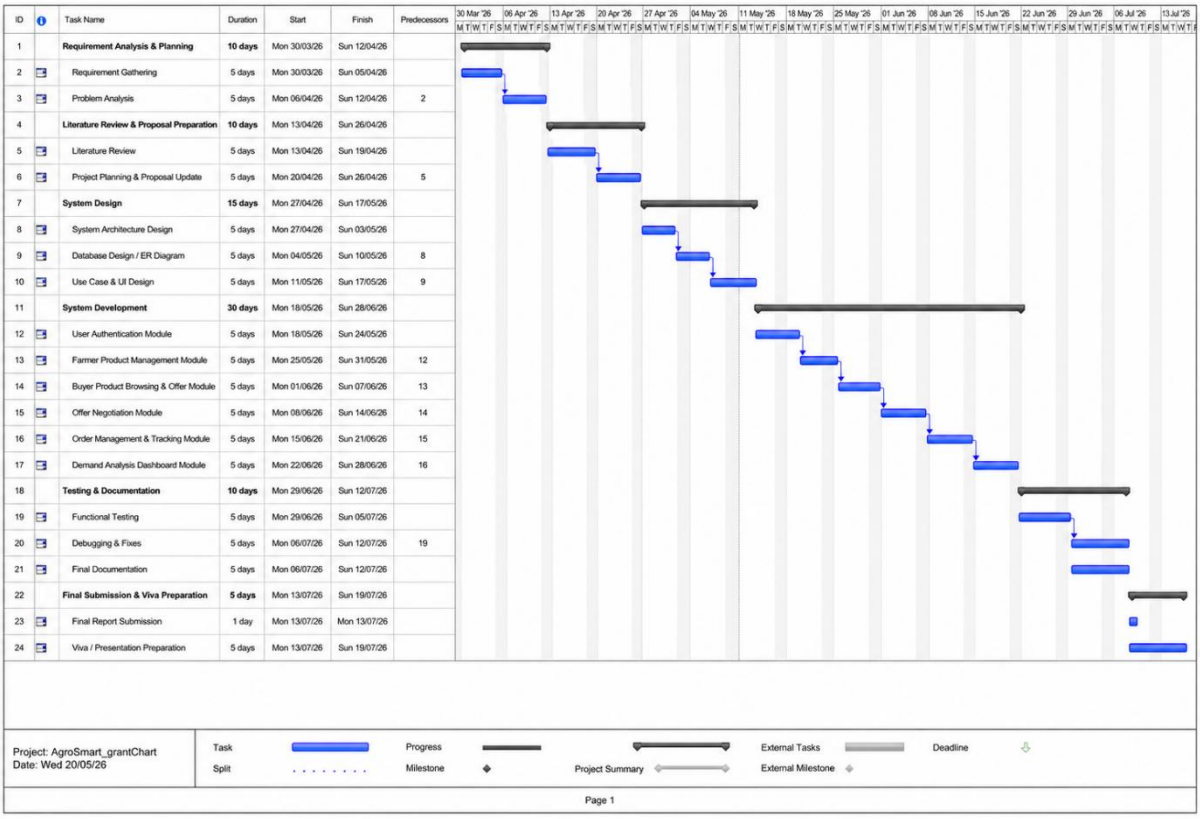
Charts and visualizations are implemented using libraries such as **Chart.js** to display demand analysis results.

5.7. Testing and Evaluation

The system is tested and evaluated to ensure:

- Functional correctness.
- User interface usability.
- Data accuracy.
- Proper functioning of negotiation and pricing logic.

6. Timeline and Milestones



ID	Task Name	Duration	Start	Finish	Predecessors
1	**Requirement Analysis & Planning**	**10 days**	Mon 30/03/26	Sun 12/04/26	
2	Requirement Gathering	5 days	Mon 30/03/26	Sun 05/04/26	
3	Problem Analysis	5 days	Mon 06/04/26	Sun 12/04/26	2
4	**Literature Review & Proposal Preparation**	**10 days**	Mon 13/04/26	Sun 26/04/26	
5	Literature Review	5 days	Mon 13/04/26	Sun 19/04/26	
6	Project Planning & Proposal Update	5 days	Mon 20/04/26	Sun 26/04/26	5
7	**System Design**	**15 days**	Mon 27/04/26	Sun 17/05/26	
8	System Architecture Design	5 days	Mon 27/04/26	Sun 03/05/26	
9	Database Design / ER Diagram	5 days	Mon 04/05/26	Sun 10/05/26	8
10	Use Case & UI Design	5 days	Mon 11/05/26	Sun 17/05/26	9
11	**System Development**	**30 days**	Mon 18/05/26	Sun 28/06/26	
12	User Authentication Module	5 days	Mon 18/05/26	Sun 24/05/26	
13	Farmer Product Management Module	5 days	Mon 25/05/26	Sun 31/05/26	12
14	Buyer Product Browsing & Offer Module	5 days	Mon 01/06/26	Sun 07/06/26	13
15	Offer Negotiation Module	5 days	Mon 08/06/26	Sun 14/06/26	14
16	Order Management & Tracking Module	5 days	Mon 15/06/26	Sun 21/06/26	15
17	Demand Analysis Dashboard Module	5 days	Mon 22/06/26	Sun 28/06/26	16
18	**Testing & Documentation**	**10 days**	Mon 29/06/26	Sun 12/07/26	
19	Functional Testing	5 days	Mon 29/06/26	Sun 05/07/26	
20	Debugging & Fixes	5 days	Mon 06/07/26	Sun 12/07/26	19
21	Final Documentation	5 days	Mon 06/07/26	Sun 12/07/26	
22	**Final Submission & Viva Preparation**	**5 days**	Mon 13/07/26	Sun 19/07/26	
23	Final Report Submission	1 day	Mon 13/07/26	Mon 13/07/26	
24	Viva / Presentation Preparation	5 days	Mon 13/07/26	Sun 19/07/26	

7. Resources and Budget

7.1. Resources

7.1.1. Hardware Requirements

- Personal Computer or Laptop (for development and testing)
- Smartphone (for testing responsiveness and usability)
- Internet connection (for development, research, and deployment)

7.1.2. Software Requirements

- Programming Languages: HTML, CSS, JavaScript, PHP
- Framework: Laravel (for backend development)
- Frontend Styling Framework: Tailwind CSS
- Database: MySQL
- Visualization Library: Chart.js
- Development Tools: Visual Studio Code
- Web Browser: Google Chrome

7.2. Budget

Item	Description	Estimated Cost (LKR)
Development Tools	Visual Studio Code, Laravel framework, Composer, NPM	0
Local Server Environment	Laravel Artisan local server	0
Database	MySQL / MySQL Workbench	0
Hardware Usage	Laptop/Desktop (existing resource)	0
Miscellaneous	Printing, binding, documentation	2,000

8. Literature Review

8.1. Existing Systems and Research

Traditional agricultural trading systems in Sri Lanka rely heavily on intermediaries, which often results in unfair pricing, reduced transparency, and lower profitability for farmers. Previous research indicates that farmers have limited control over pricing decisions, as intermediaries dominate the trading process and capture a significant share of the profit margin. To overcome these challenges, digital agriculture platforms have been introduced in Sri Lanka to facilitate direct interactions between farmers and buyers. These platforms improve product visibility and accessibility by enabling online trading. However, most existing systems primarily focus on basic buying and selling functionalities and do not support structured price negotiation between farmers and buyers.

Furthermore, many current systems lack the capability to collect and analyze buyer activities such as product searches, views, and offers, which are essential for understanding real market demand. In addition, the absence of integrated feedback or review mechanisms reduces transparency and trust within these platforms. Additionally, many existing systems do not provide structured mechanisms for farmers to publish products and receive competitive price offers from multiple buyers, limiting effective price determination.

8.2. Smart Pricing and Data-Driven Approaches

Recent studies highlight the importance of data-driven decision-making in modern agricultural systems. Smart pricing mechanisms, even when implemented using rule-based algorithms, can assist farmers in setting competitive and fair prices based on market trends and demand levels.

While advanced systems utilize machine learning techniques for price prediction, such approaches often require large datasets and significant computational resources. Therefore, for small-scale and developing systems, rule-based pricing methods provide a practical and cost-effective alternative for supporting informed pricing decisions.

8.3. Demand Analysis in Agriculture

Demand analysis plays a critical role in maintaining balance between supply and demand in agricultural markets. Research shows that the absence of accurate demand data leads to inefficiencies such as oversupply or shortages of crops, which directly affect price stability.

By analyzing buyer activities such as product searches, views, offers, and completed orders, digital platforms can identify demand patterns and generate valuable insights. These insights can be presented through dashboards to support farmers in making better production, pricing, and marketing decisions.

8.4. Research Gap

Although several digital agriculture platforms exist, there is a lack of integrated systems that combine direct farmer-to-buyer trading, smart pricing support, demand analysis based on buyer activities, structured negotiation mechanisms, location-based matching, and feedback systems within a single platform. Additionally, many existing systems do not adequately consider local market conditions or provide simple, user-friendly, and cost-effective solutions suitable for agricultural users in Sri Lanka.

9. References

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